CDS540 Group Project

Timeline

• Form your group and report the group members to the TA. (Oct 9)

• Select a Computer Vision project topic and report it to the TA. (Oct 16)

• Submit your group presentation slides and report to Moodle. (Nov 20)

• Deliver your group presentation according to the announced schedule. (Nov 20 & 27)

• Complete and submit your peer evaluation form privately. (Nov 20 & Nov 27)

Candidate Topics

Your project should be related to **Computer Vision**. The following topics are examples; other Computer Vision topics are also welcome.

1. Image processing and enhancement

Image denoising, deblurring, contrast enhancement, super-resolution, or image restoration.

2. Image classification and recognition

Object or scene classification, visual recognition, or visual anomaly detection.

3. Object detection and image segmentation

Object detection, semantic or instance segmentation, text detection, or medical image segmentation.

4. Video understanding and analysis

Object tracking, action recognition, motion estimation, or video event detection.

5. 3D computer vision

Depth estimation, stereo vision, 3D reconstruction, or point cloud processing.

6. Generative models for visual content

Image or video generation, image editing, image-to-image translation, or style transfer.

Notes for Selecting Your Topic

Personal Interests

Choose a visual task that interests your team. For example, a photography-related project could enhance image quality, while a sports-related project could track players or recognize actions in videos.

Course Content and Learning Goals

Apply concepts covered in CDS540, such as image processing, convolutional neural networks, vision transformers, detection, segmentation, or visual generation. Topics beyond the course content are welcome as long as they are related to Computer Vision.

Practicality and Feasibility

Consider the available data, computing resources, time, and your team's experience. You may use existing libraries and pre-trained models. Choose a manageable task and clearly explain your implementation, experiments, and findings.

Relevance and Impact

Consider a practical visual problem, such as detecting product defects, analyzing traffic, or improving low-quality images. Define a clear objective and evaluate whether your approach achieves it.

Team Collaboration

Choose a topic that reflects the interests and skills of all group members. Divide the work fairly so that each member makes a meaningful contribution.

Notes for the Group Project Report

Upload the report to Moodle as a **.pdf** file. Use **Times New Roman, 12-point font, 1.25 line spacing, and 0.5-line spacing after each paragraph**. The main text must not exceed **8 pages** of A4 paper in portrait orientation.

Describe the problem, relevant background, method, data, experiments, results, and limitations. Acknowledge any external code, datasets, pre-trained models, and other sources used.

Notes for Group Presentation Slides

Upload the slides to Moodle as a **.ppt, .pptx, or .pdf** file. You must use the slides downloaded from Moodle when presenting.

Notes for the Group Presentation

Each group has **20 minutes** in total: **17 minutes for the presentation and 3 minutes for Q&A**. The presentation order will be random and announced on Moodle before the presentations begin.
Please note: If you exceed the time limit, your presentation will be interrupted.

Notes for Peer Evaluation

Individual scores will be based on group performance and peer evaluation. Each student must privately submit an evaluation of the contributions and performance of the other members of their group.